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(54) BATTERY AND BATTERY SYSTEM INCLUDING PRESSURE VESSEL

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H01M 50/152 (2021.01)

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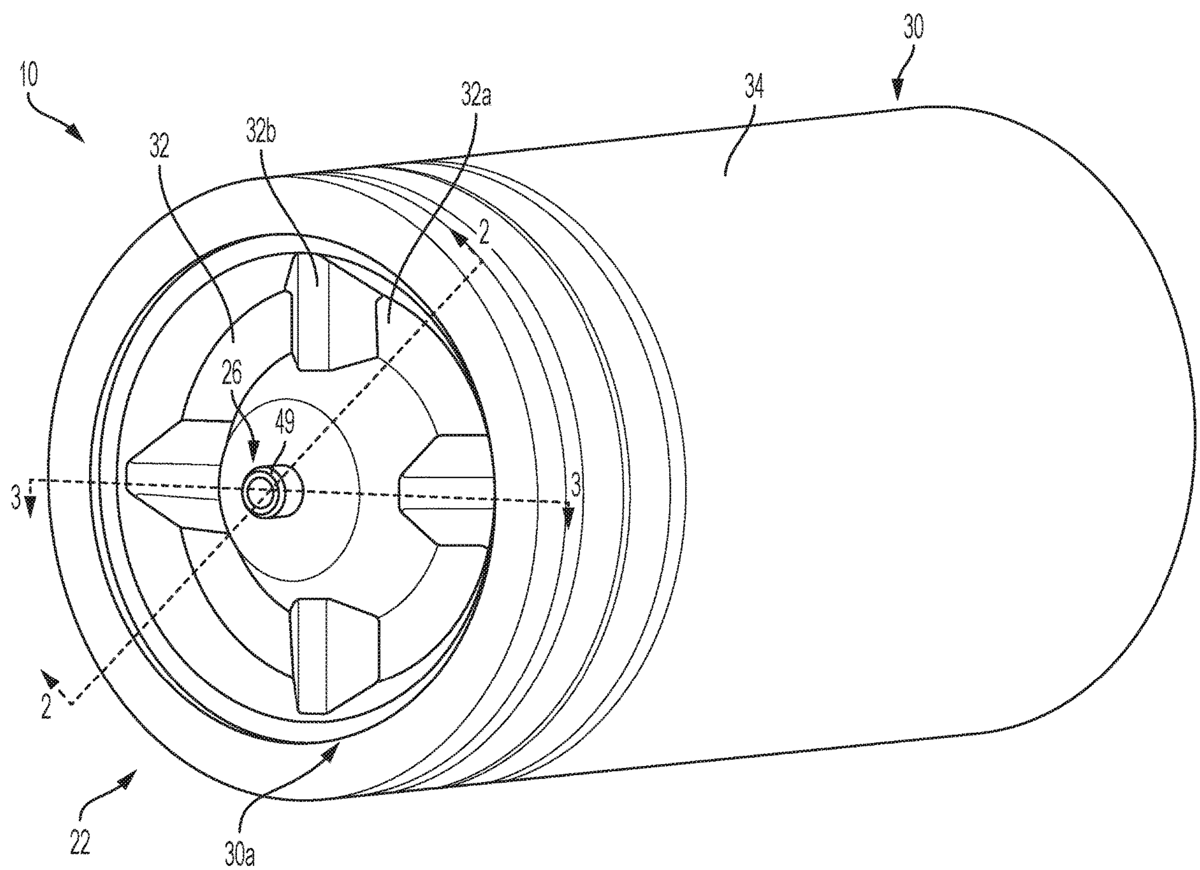
(57) **ABSTRACT**

A battery includes a tubular battery cell including an anode, a cathode and a solid electrolyte. The battery cell defines a central through hole. The battery further includes a container. The battery cell is inside of the container. The container includes a fluid inlet configured to provide fluid to the central through hole and a fluid outlet configured to receive fluid from the central through hole.



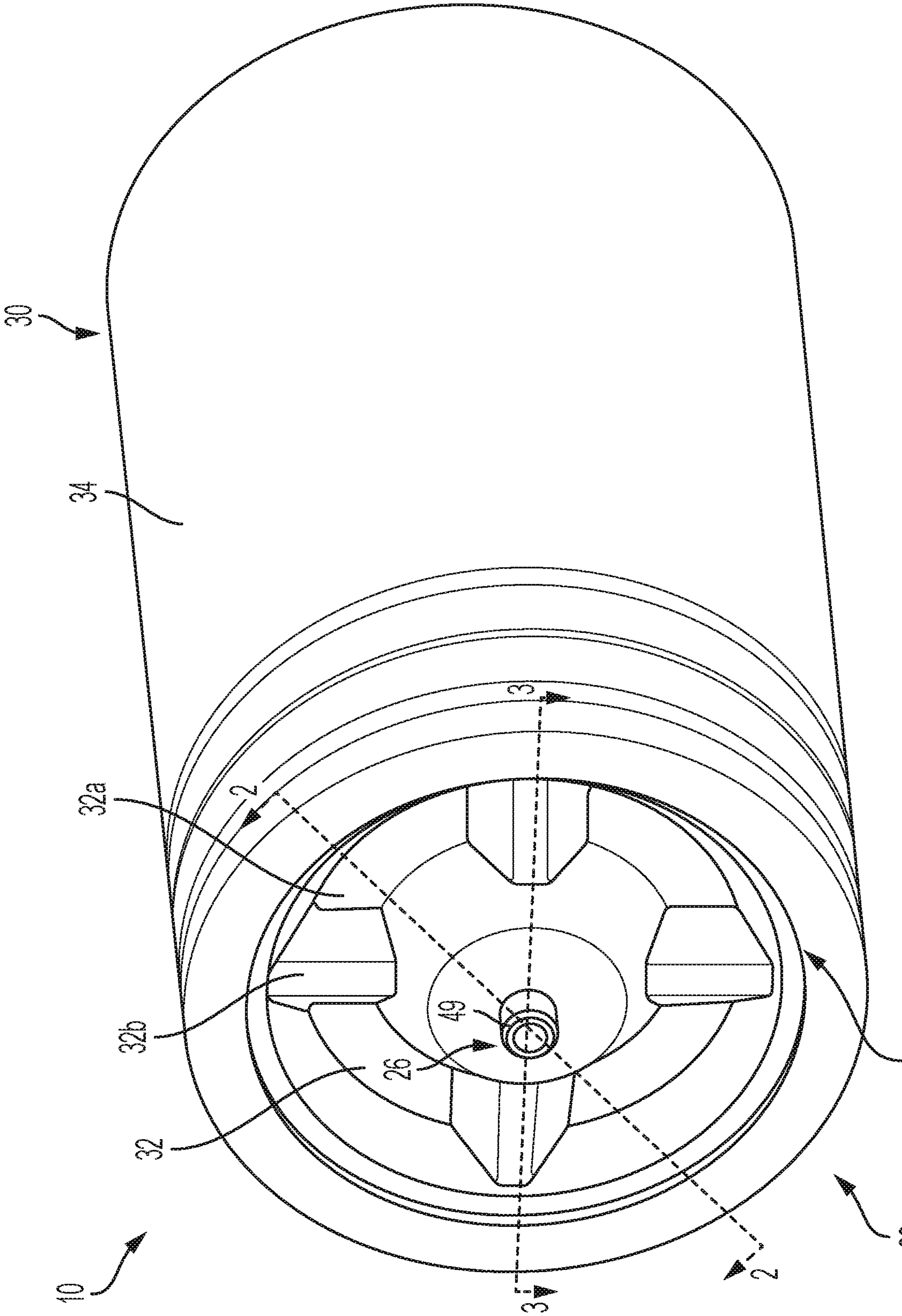
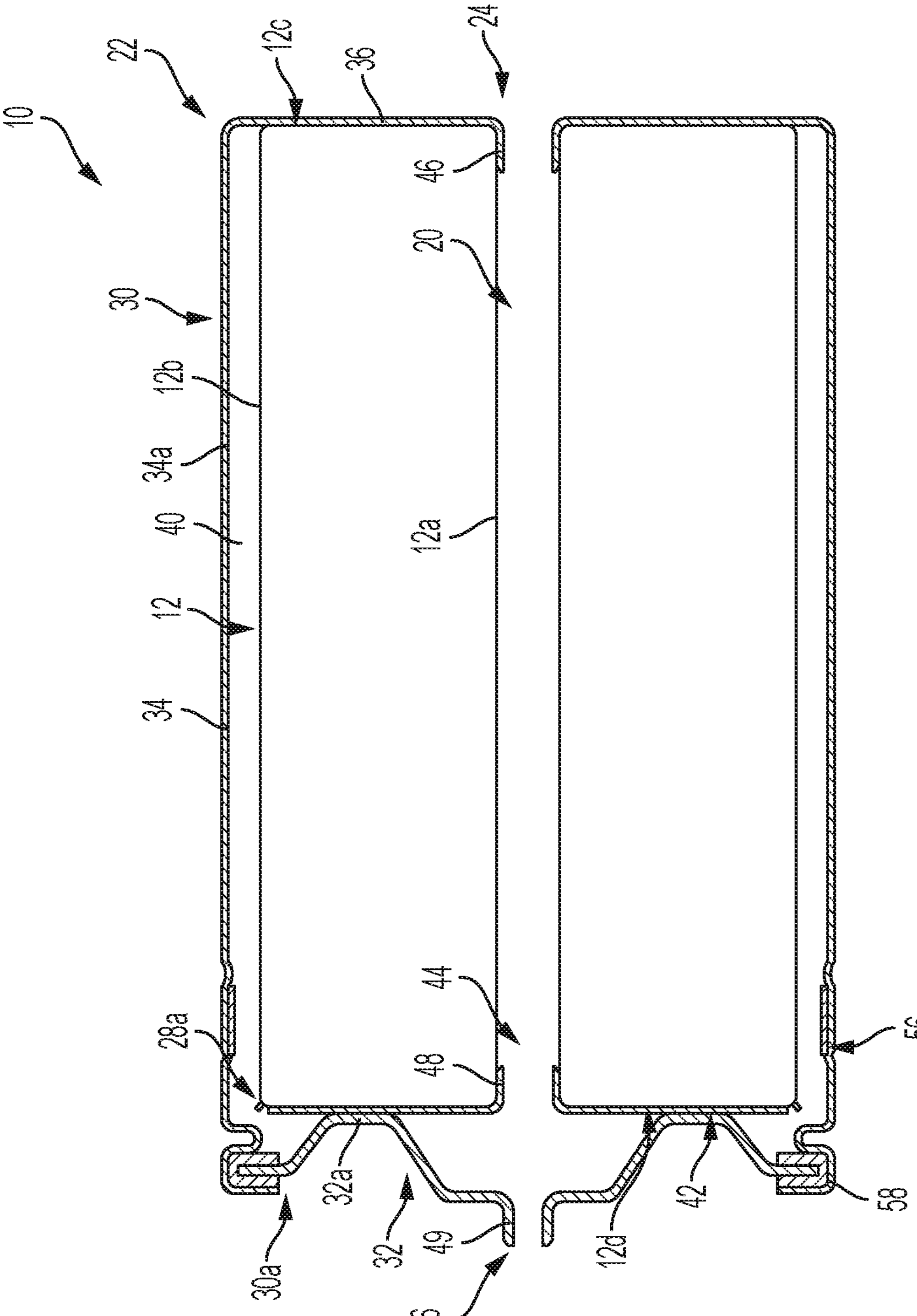


FIG. 1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12c
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30
12b
34a
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12a
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28a
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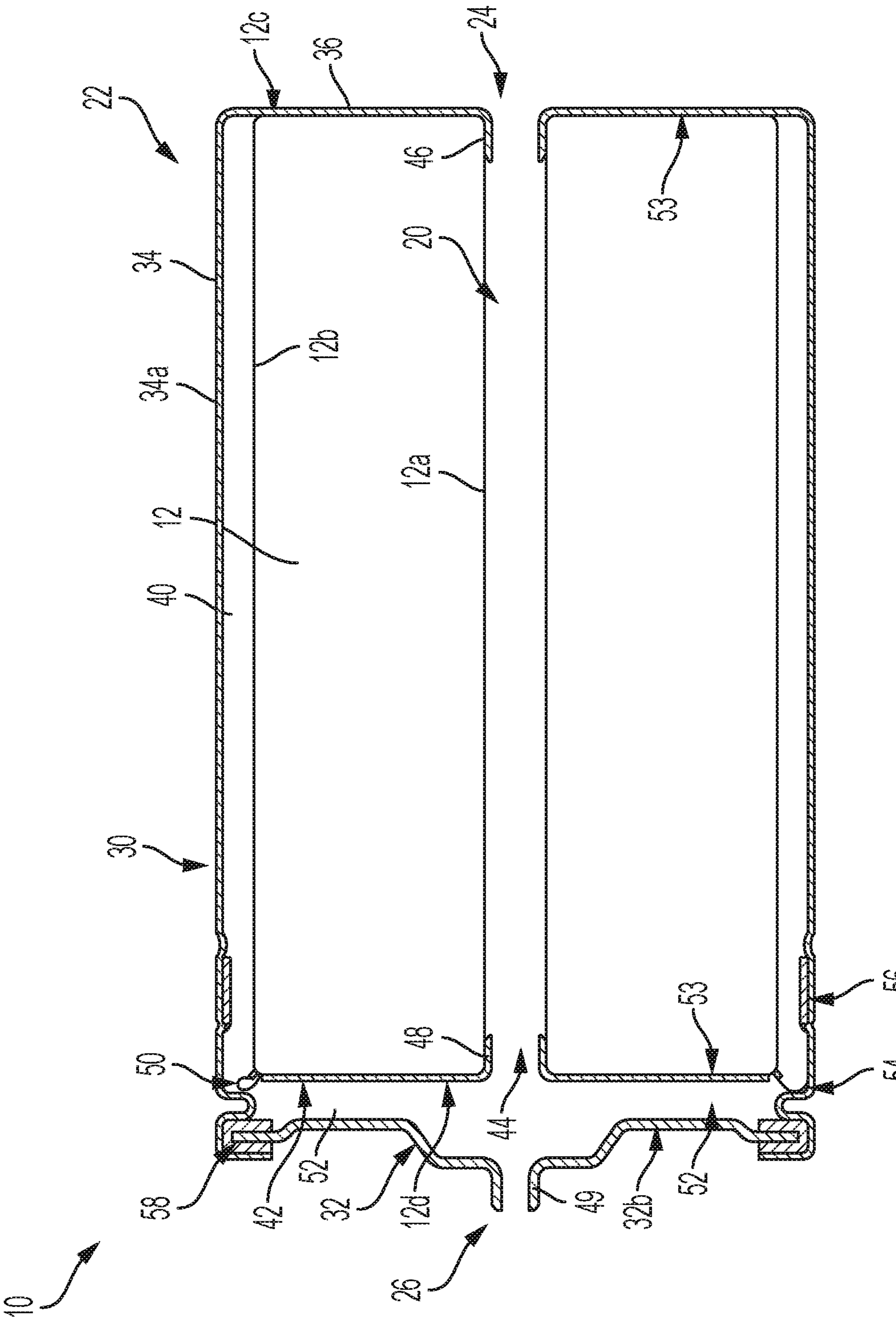
56
32a
42
58
30a
32
49
12d
26

FIG. 2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44
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32
12d
26
49
32b
52

FIG. 3

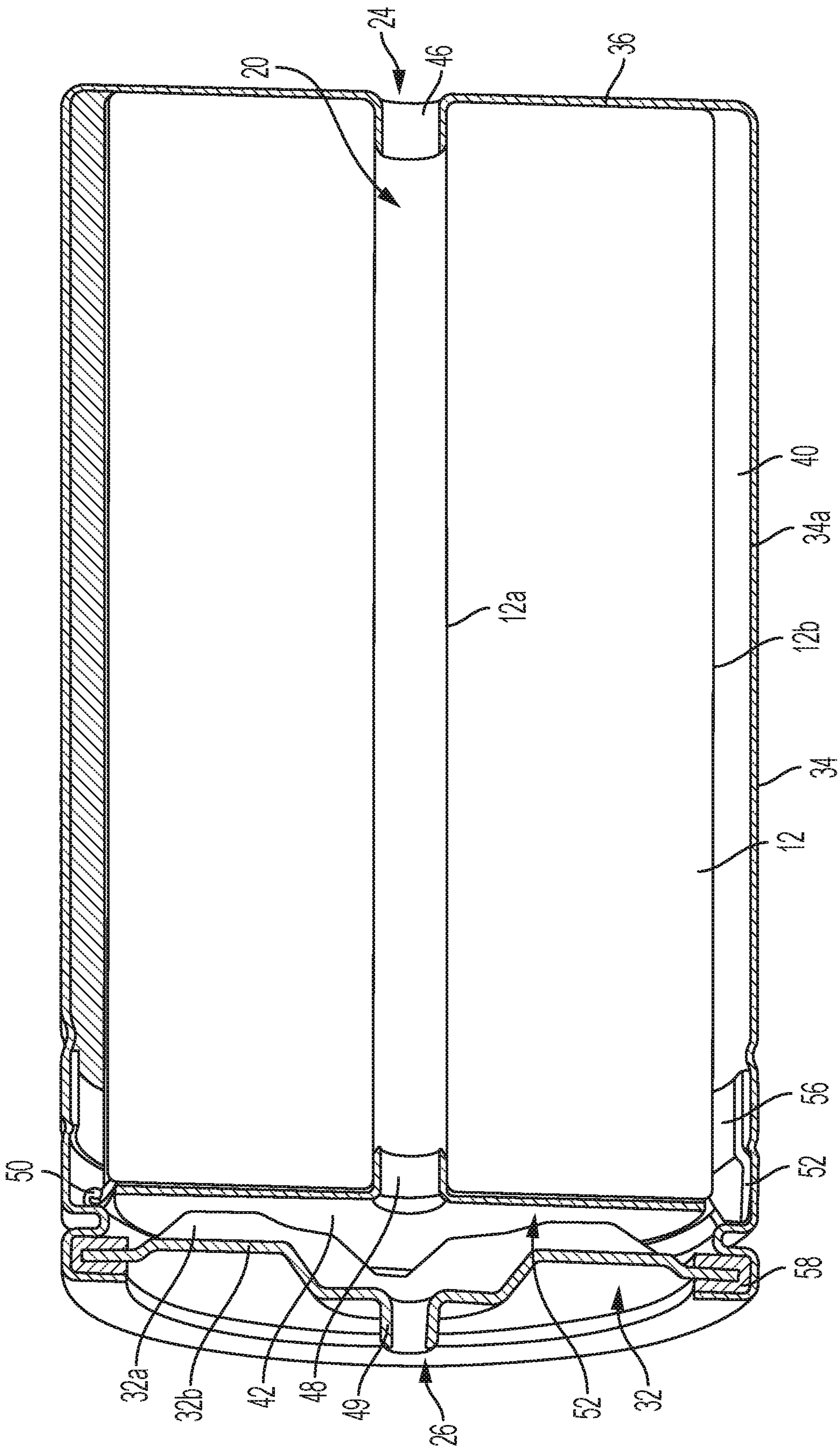


FIG. 4

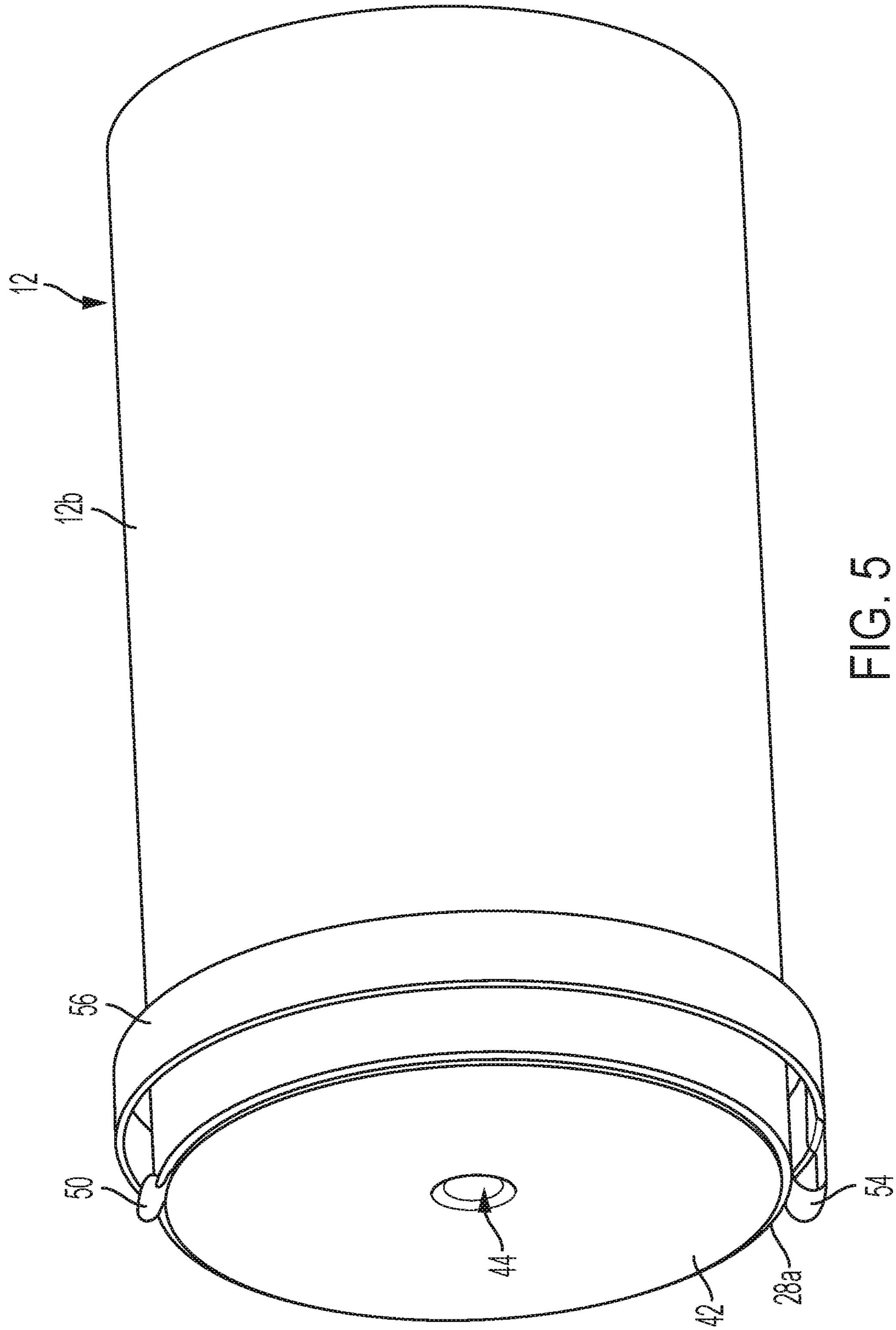


FIG. 5

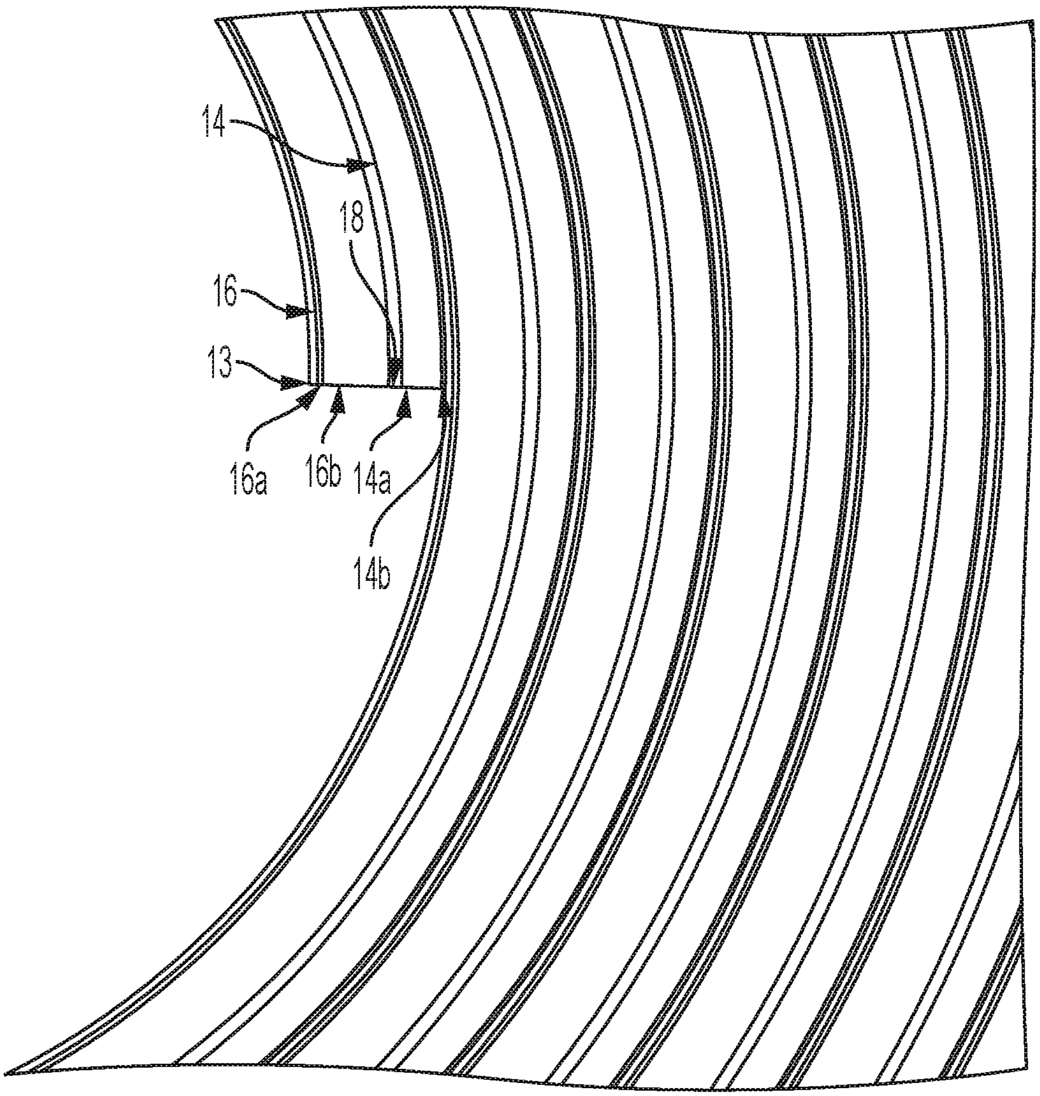


FIG. 6

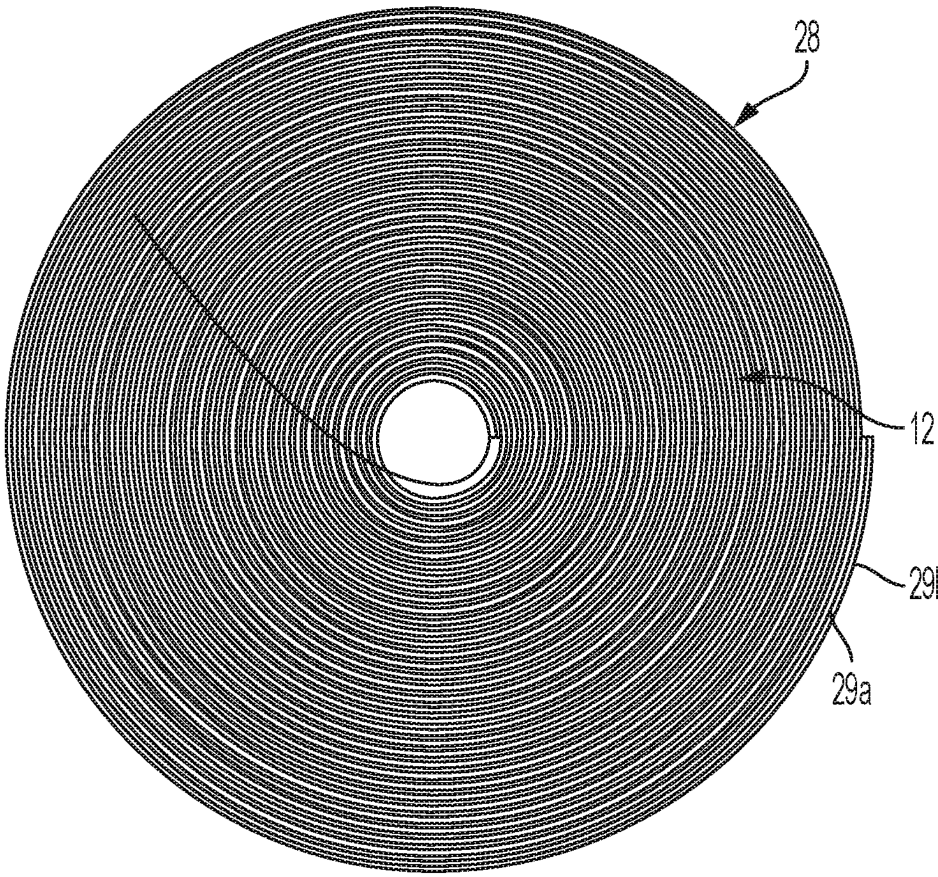
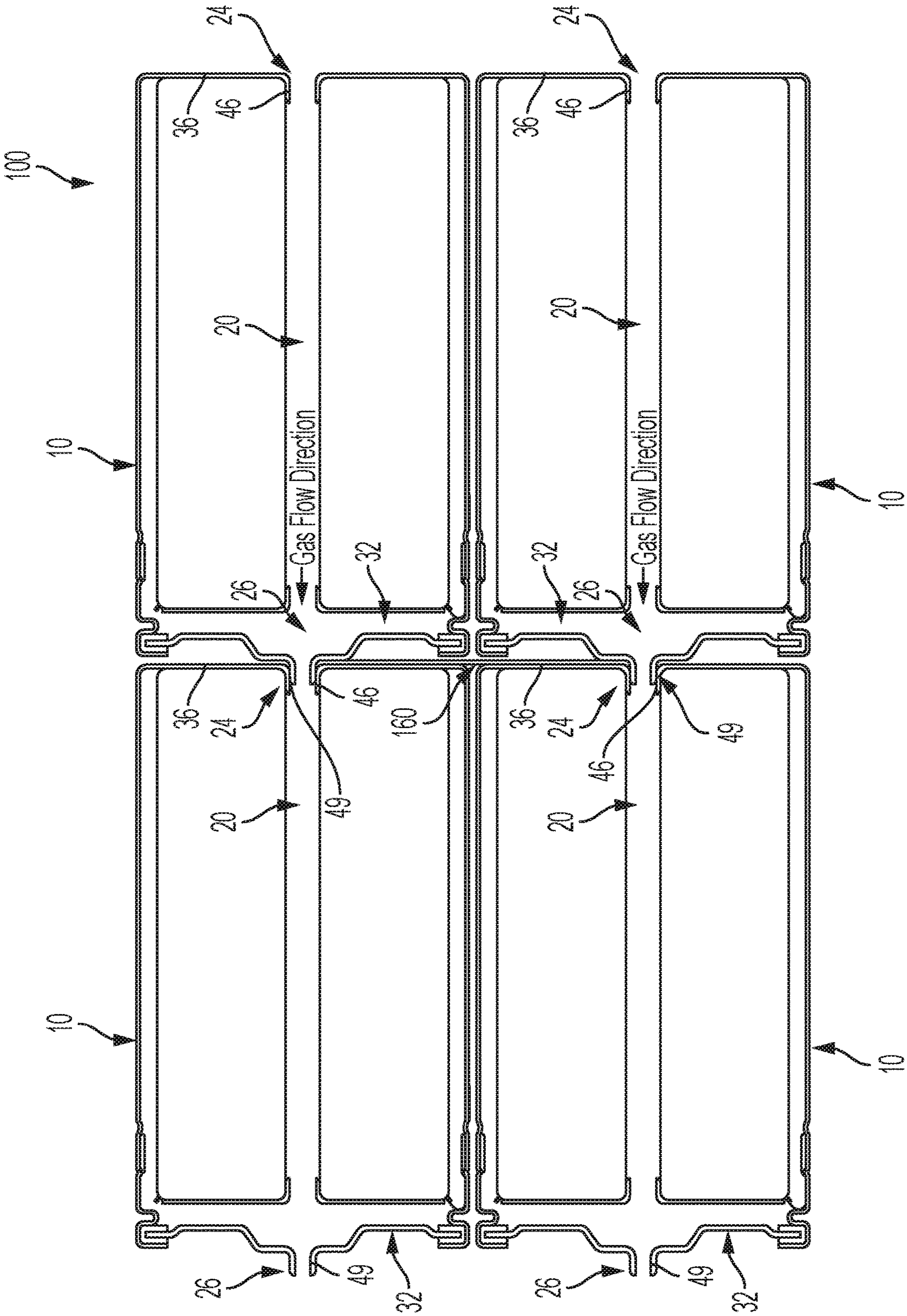


FIG. 7


100
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10
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20
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24
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26
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32
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36
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36
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46
46
49
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49
49
160
Gas Flow Direction
Gas Flow Direction

FIG. 8

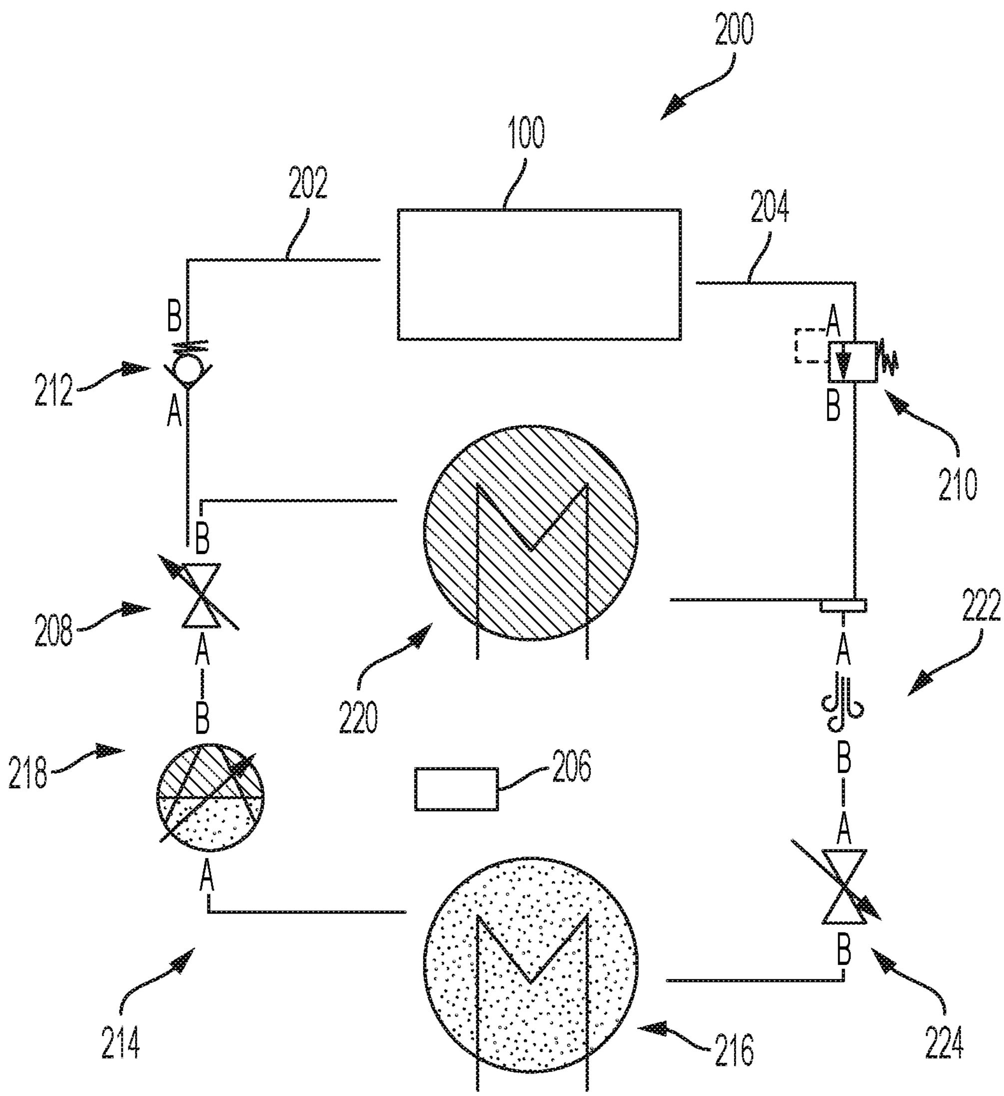
200
100
202
204
B
A
212
A
B
210
B
208
A
B
220
222
A
B
A
218
206
A
214
216
B
224

FIG. 9

BATTERY AND BATTERY SYSTEM INCLUDING PRESSURE VESSEL

TECHNICAL FIELD

[0001] The present disclosure relates generally to solid state electrolyte batteries.

BACKGROUND

[0002] Solid state electrolyte battery cells require a different environment than liquid electrolyte batteries. Accordingly, there is a need for an improved battery design.

SUMMARY

[0003] A battery includes a tubular battery cell including an anode, a cathode and a solid electrolyte. The battery cell defines a central through hole. The battery further includes a container. The battery cell is inside of the container. The container includes a fluid inlet configured to provide fluid to the central through hole and a fluid outlet configured to receive fluid from the central through hole.

[0004] In examples, the battery cell has a jelly roll configuration.

[0005] In some aspects, the techniques described herein relate to a battery further including a flexible housing inside the container surrounding the battery cell, the flexible housing configured for allowing the battery cell to radially expand and contract within the flexible housing.

[0006] In examples, the flexible housing is a bag an internal polypropylene shell formed of a first polymer and an external shell formed of a second polymer.

[0007] In examples, the flexible housing is oversized with respect to the battery cell to allow for up to 10% expansion of the battery cell during charge and/or discharge of the battery cell.

[0008] In examples, an outer circumferential surface of the battery cell and an inner circumferential surface of a tubular section of the container define a tubular space therebetween to allow gas to flow outside of the outer circumferential surface of the battery cell to generate a radially inward force on the outer circumferential surface of the battery cell.

[0009] In examples, the battery further includes an internal retainer inside the container and fixed to a first longitudinal end of the battery cell, the internal retainer including a central hole therein aligned with the central through hole of the battery cell to cause fluid flowing from the fluid inlet to the fluid outlet to pass through the central hole of the internal retainer.

[0010] In examples, the container includes a can and a lid fixed to the can, the can including the tubular section, the internal retainer being physically connected to the lid and held in place within the can by the lid.

[0011] In examples, a radially extending section of the can defines a first longitudinal end of the container and the lid defines a second longitudinal end of the container, the lid including the fluid outlet, the radially extending section of the can including the fluid inlet.

[0012] In examples, the radially extending section of the can includes an axially extending centering part protruding into the central through hole of the battery cell at a first longitudinal end of the battery cell.

[0013] In examples, the internal retainer includes an axially extending centering part protruding into the central through hole of the battery cell at a second longitudinal end of the battery cell.

[0014] In examples, the internal retainer is electrically connected to the lid and electrically connects a first lead of the battery cell to the lid.

[0015] In examples, the can is electrically connected to a second lead of the battery cell.

[0016] In examples, the first lead is positive and the second lead is negative.

[0017] In examples, the lid is configured to provide fluid flow channels between the lid and the internal retainer for fluid to pass from the central through hole of the battery cell to the tubular space.

[0018] A battery assembly is also provided that includes at least two of the batteries. At least one of the batteries being an upstream battery with respect to a fluid flow and at least one of the batteries being a downstream battery with respect to the fluid flow. The central through holes of the at least upstream battery and the at least one downstream battery are aligned with each other and the fluid inlet of one of the at least one downstream battery is connected to the fluid outlet of the at least upstream battery.

[0019] A battery system is also provided including the battery assembly; and a pressurization system for conveying compressed gas into the batteries to generate a force on the battery cells for maintaining uniform pressure on each battery cell.

[0020] A method of constructing a battery is also provided. The method includes inserting a tubular battery cell into an open end of a can such a tubular space is present between an outer circumferential surface of the battery cell and an inner circumferential surface of the can, a first longitudinal end of the battery cell facing a radially extending end section of the can, the radially extending end section of the can including a fluid inlet configured to provide fluid to a central through hole of the battery cell, the battery cell including an anode, a cathode and a solid electrolyte; installing an internal retainer onto a second longitudinal end of the battery cell; and connecting a lid to the open end of the can such that the lid is electrically connected to the internal retainer during operation of the battery, the lid including a fluid outlet configured to receive fluid from the central through hole.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The present disclosure is described below by reference to the following drawings, in which:

[0022] FIG. **1** shows a perspective view of a battery according to the present disclosure;

[0023] FIG. **2** shows a cross-sectional view of the battery along **2-2** in FIG. **1**;

[0024] FIG. **3** shows a cross-sectional view of the battery along **3-3** in FIG. **1**;

[0025] FIG. **4** shows a cross-sectional perspective view of the battery of FIG. **1**;

[0026] FIG. **5** shows a perspective view of the battery shown in FIG. **1** with the container omitted;

[0027] FIG. **6** shows a partial cross-sectional view of a tubular battery cell having a jelly roll configuration;

[0028] FIG. **7** shows the tubular battery cell of FIG. **6** inside a flexible housing;

[0029] FIG. **8** schematically shows a battery assembly including multiple batteries connected together;

[0030] FIG. **9** shows a system for providing compressed and heated fluid to the battery assembly of FIG. **6**.

DETAILED DESCRIPTION

[0031] Solid-state battery cells can require large pressures to maintain contact between the active materials of the anode and cathode and the electrolyte so that ion exchange is possible with the required contact area. Additionally, higher operating temperatures can also be also required. The present disclosure provides a battery and a battery system in which a battery cell can be compressed and heated during operation. The present disclosure allows compressed and heated CO_2 gas, to apply pressure and temperature to a tubular battery cell by allowing the gas around the battery cell to compress the battery cell in a controlled format. As such, the interior design of the battery includes fluid flow chambers that allow the gas to heat and pressurize the tubular battery cell so as to maintain even compressive pressure through the tubular battery cell. Furthermore, the battery is designed so that a system can pressurize and flow gas from one battery to the next using a series connection for the gas and electrical connections.

[0032] FIGS. **1** to **7** show different views of a battery **10**. Battery **10** includes a tubular battery cell **12** including an anode **14**, a cathode **16** and a solid electrolyte **18** (FIG. **6**). The battery cell **12** defines a central through hole **20**. Battery **10** further includes a container **22**, and the battery cell **12** is inside of the container **22**. The container **22** includes a fluid inlet **24** configured to provide fluid to the central through hole **20** and a fluid outlet **26** configured to receive fluid from the central through hole **20**. Hole **20** is defined by an inner circumferential surface **12***a* of battery cell **12**.

[0033] As illustrated in FIG. **6**, battery cell **12** has a jelly roll configuration, including a strip of battery material rolled into a tubular form. The strip of battery material includes a plurality of layers, each extending an entirety of the length of the strip. Defining an inner circumferential surface of battery cell **12** is a separator material **13**, with cathode **16** contacting an outward facing surface of separator material **13**. Cathode **16** is formed by a cathode current collector foil **16***a* that directly contacts an outward facing surface of separator material **13** and a cathode material **16***b* directly contacting an outward facing surface of cathode current collector foil **16***a*. Next, solid electrolyte **18** directly contacts an outward facing surface of cathode material **16***b*, and then anode **14** contact an outward facing surface of solid electrolyte **18**. Anode **14** is formed by an anode material **14***a* that directly contacts an outward facing surface of solid electrolyte **18** and an anode current collector foil **14***b* directly contacting an outward facing surface of anode material **14***a*.

[0034] The solid electrolyte **18** can be for example formed of a flexible material, which can for example be a polymer. The anode material **14***a* can be for example formed of graphite, silicon graphite mix, lithium metal, or lithium-titanium-oxide. The cathode material **16***b* can be for example formed of lithium nickel cobalt manganese oxide (NMC), lithium iron phosphate (LFP), lithium manganese oxide (LMO).

[0035] As discussed further below, the fluid can be a gas. In one example, the gas is compressed CO_2.

[0036] As illustrated in FIG. **7**, battery **10** further includes a flexible housing **28** inside the container **12** surrounding the battery cell **12**. The flexible housing **28** is configured to allow the battery cell **12** to radially expand and contract within the flexible housing **28**. The flexible housing **28** also provides protection from the gas for the active materials of cell **12** as well as electrical isolation from the container **22**. The flexible housing **28** can be a bag which has an internal shell **29***a* formed of a first polymer and an external shell **29***b* formed of a second polymer. Housing **28** can be sealed at a seam **28***a* on an outer circumferential corner of housing **28**. The first polymer can be a polypropylene and the second polymer can be a polyimide, a polycarbonate, or a polysulfone. The flexible housing **28** can be oversized with respect to the battery cell **12** to allow for up to 10% expansion of the battery cell **12** during charge and/or discharge of the battery cell **12**.

[0037] Container **22** can include a can **30** including an open end **30***a* and a lid **32** closing the open end **30***a*. Can **30** includes a tubular section **34** and a radially extending section **36**. Radially extending section **36** closes a first longitudinal end of the tubular section **34** and lid **32** closes a second longitudinal end of the tubular section **34**. Radially extending section **36** includes fluid inlet **24** and lid **32** includes fluid outlet **26**. Further, a first longitudinal end **12***c* of the battery cell **12** is fixed to radially extending section **36**.

[0038] An outer circumferential surface **12***b* of battery cell **12**, more specifically of flexible housing **28**, and an inner circumferential surface **34***a* of tubular section **34** defines a tubular space **40** therebetween to allow gas to flow outside of the outer circumferential surface **12***b* to generate a radially inward force on the outer circumferential surface **12***b*.

[0039] Battery **10** further includes an internal retainer **42** inside the container **12** that is fixed to a second longitudinal end **12***d* of the battery cell **12**. The internal retainer **42** includes a central hole **44** therein aligned with the central through hole **20** of the battery cell **12** to cause fluid flowing from the fluid inlet **24** to the fluid outlet **26** to pass through the central hole **44** of the internal retainer **42**. The internal retainer **42** is physically connected to the lid **32** and held in place within the can **30** by the lid **32**.

[0040] The radially extending section **36** of the can **30** includes an axially extending centering part **46** protruding into the central through hole **20** of the battery cell **12** at a first longitudinal end of the battery cell **12**. Similarly, the internal retainer **42** includes an axially extending centering part **48** protruding into the central through hole **20** of the battery cell **12** at a second longitudinal end of the battery cell **12**. Centering parts **46**, **48** thus act on opposite longitudinal ends of hole **20** to center battery cell **12** within container **22**. Each of radially extending section **36** and internal retainer **42** are formed as a disc with a tubular extension in the form of the respective centering part **46**, **48** protruding from the disc into the hole **20**. Axially extending centering part **46** defines fluid inlet **24** and axially extending centering part **48** defines central hole **44**. Lid **34** also includes an axially extending centering part **49** protruding away from hole **20**. Axially extending centering part **49** defines fluid outlet **26** and can be inserted into the fluid inlet **24** of another battery **10** as discussed further below.

[0041] The internal retainer **42** is electrically connected to the lid **32** and electrically connects a first lead **50** of the battery cell **12** to the lid **32**. First lead **50** can be connected to internal retainer **42** and lid **32** by brazing, laser/ultrasonic welding, or clinching. The first lead **50** can be positive and can be formed by a cable including a wire. First lead **50**

extends from the cathode **16** through the seam **28**_a_ of housing **28** to the retainer **42**. As shown in FIG. **1**, the lid **32** includes recessed sections **32**_a_ and raised sections **32**_b_ that alternate with each other. Recessed sections **32**_a_ define contact sections that contact internal retainer **42** to physically and electrically connect lid **32** to internal retainer **42**. Raised sections **32**_b_ define fluid flow channels **52** within container **22** for fluid to flow from hole **20** to tubular space **40**. First longitudinal end **12**_c_ of battery cell **12** is sealingly fixed to radially extending section **36** of can **30** by adhesive **53** and second longitudinal end **12**_d_ of battery cell **12** is sealingly fixed to internal retainer **42** by adhesive **53** to prevent fluid from flowing between first longitudinal end **12**_c_ and fluid outlet **26** or between second longitudinal end **12**_d_ and internal retainer **42** so that gas pressure cannot be used to compress the battery cell **12** in the axial direction. The fluid receiving spaces of battery **10** provide a chamber built into the container **22**, such that a static pressure of the fluid reacts against both the inner circumferential surface **12**_a_ and outer circumferential surface **12**_b_ of battery cell **12** to form a uniform pressure on the cell **12**, which promotes proper solid-state cell function.

[0042] The can **30** is electrically connected to a second lead **54** of the battery cell **12**. The second lead **54** can be negative and can be formed by a cable including a wire. Second lead **54** extends from the anode **14** through the seam **28**_a_ of the housing **28** to an electrically conductive ring **56**, which is connected to inner circumferential surface **34**_a_ of tubular section **34** and electrically connects second lead **54** to can **30**. Second lead **54** can be connected to ring **56** by brazing, laser/ultrasonic welding, soldering, projection welding, spot welding or clinching. The electrically conductive ring **56** can be attached to the can **30** and retained using a rolling process to lock the ring **56** into place around the perimeter.

[0043] Lid **32** and can **30** are electrically insulated from each other by a flexible seal **58** surrounding an outer circumference of lid **32**. Lid **32** can define a positive terminal of battery **10** and radially extending section **36** of can **30** can define a negative terminal of battery **10**.

[0044] A method of constructing a battery **10** including inserting a tubular battery cell **12** into an open end **30**_a_ of a can **30** such a tubular space **40** is present between an outer circumferential surface **12**_b_ of the battery cell **12** and an inner circumferential surface **34**_a_ of the can **30** and a first longitudinal end **12**_c_ of the battery cell **12** faces radially extending end section **36** of the can **30**. The method of constructing a battery **10** also includes installing the internal retainer **42** onto a second longitudinal end **12**_d_ of the battery cell **12** and connecting a lid **32** to the open end of the can **30** such that the lid **32** is electrically connected to the internal retainer **42** during operation of the battery **10**.

[0045] FIG. **6** shows a battery assembly **100** including a plurality of batteries **10** connected together. The central through holes **28** of the at least two batteries **10** are aligned with each other and the fluid inlet **24** of at least one of the one of the batteries **10** is connected to the fluid outlet **26** of another of the batteries **10**. In the example shown in FIG. **6**, four batteries **10** are shown, but it should be understood that such a battery system can include tens or hundreds of batteries **10**. FIG. **6** shows two sets of batteries **10**, with the two batteries in the top of the figure forming a first set and the two batteries in the bottom of the figure forming a second set. Each set includes two batteries **10** arranged in series both from a fluid flow relationship and an electrical relationship. Each of the upstream batteries **10** on the right side of FIG. **6** are aligned with one of the downstream batteries **10** on the left side of FIG. **6**.

[0046] For each battery set, the fluid outlet **26** of one battery **10** is directly connected to the fluid inlet **24** of the other battery **10** and the lid **32** of one battery **10** is engaged with the radially extending section **36** of the other battery. More specifically, for each battery set, the axially extending centering part **49** of the lid **32** of one battery **10** is inserted into the axially extending centering part **46** of radially extending section **36** of the other battery **10**. Thus, fluid can flow into an inlet **24** of the first battery **10**, through the hole **20** of the first battery **10**, and then out of the fluid outlet **26** of the first battery directly into inlet **24** of the second battery **10**, through the hole **20** of the second battery **10**, and then out of the fluid outlet **26** of the second battery.

[0047] The two sets of batteries can also be electrically connected together in parallel by a parallel connection **160**. The parallel connection **160** contacts the positive terminals of the batteries on the right side of the figure and the negative terminals of the batteries on the right side of the terminal to electrically connect the first and second sets in parallel.

[0048] FIG. **7** shows a system **200** for providing compressed and heated fluid to the battery assembly **100**. System **200** is configured for conveying heated compressed gas into the batteries **10** of battery assembly **100** to provide constant pressure to batteries **10** during the charging and discharging in order to provide sufficient ionic conductivity between the anode material and the cathode material, and to maintain the operating temperature within the range of acceptable operating temperatures.

[0049] System **200** is configured for conveying compressed gas into batteries **10** via the fluid inlet **24** to generate a force on the cells **12** for maintaining uniform pressure on each battery cell **12**. The system **200** includes a gas inlet line **202** feeding into the fluid inlets **24** of upstream batteries **10** and a gas outlet line **204** for gas exiting the fluid outlets **26** of the downstream batteries **10**.

[0050] System **200** further includes a controller **206** configured to control a flow rate of heated compressed gas to batteries **10** to control the temperature within pressure batteries **10** and configured to control a pressure within the batteries **10** to maintain uniform pressure on each battery **10**. More specifically, system **200** includes a control valve **208** in gas inlet line **202** and a pressure relief valve **210** in gas outlet line **204**. Controller **206** is configured to control a flow rate of heated compressed gas to batteries **10** by sending control signals to control valve **208** and to control the pressure within the batteries **10** by sending control signals to pressure relief valve **210**. Pressure relief valve **210** also maintains the pressure of the batteries **10** when the vehicle is shut off. The system **200** includes a check valve **212** in the gas inlet line **202** for allowing compressed gas to flow into the fluid inlets **24** of upstream batteries **10** and preventing backflow from the fluid inlets **24** of upstream batteries **10**.

[0051] System **200** can be onboard a motor vehicle and further include an HVAC system **214** configured for absorbing heat from a cabin of the motor vehicle and producing heated compressed gas that is supplied to system **200**. The heated compressed gas can advantageously be carbon dioxide. The HVAC system **214** includes an evaporator **216** within the cabin of the motor vehicle configured for absorbing heat from the cabin and evaporating liquid refrigerant into gas. HVAC system **214** further includes a compressor **218** configured for compressing and heating the gas, and a condenser **220** configured for cooling the heated compressed output by the compressor and condensing the gas into a liquid.

[0052] The control valve **208** is provided directly downstream of compressor **218** and is configured for directing the heat compressed output by the compressor **218** to fluid inlets **24** of the upstream batteries **10** and/or to the condenser **220**. The heated compressed gas can thus be directed to fluid inlets **24** of the upstream batteries **10** through the check valve **52** and into batteries **10** to heat the battery cells **12** and pressurize batteries **10** to maintain sufficient contact between the battery cells **12**.

[0053] The gas exiting from the fluid outlets **26** of the downstream batteries **10** of assembly **100** flows through pressure relief valve **210** to merge with the liquid refrigerant output by condenser **220**. The HVAC system **214** further includes a dryer **222** downstream from the pressure relief valve and configured for receiving liquid refrigerant from condenser **220**, which has merged with the gas exiting from the fluid outlets **26** of the downstream batteries **10** of assembly **100**. An expansion valve **224** of HVAC system **214** is downstream of the dryer **222** for regulating the flow of liquid refrigerant into evaporator **216**.

LIST OF REFERENCE CHARACTERS

[0054] **10** batteries
[0055] **12** battery cell
[0056] **12***a* inner circumferential surface
[0057] **12***b* outer circumferential surface
[0058] **12***c* first longitudinal end
[0059] **12***d* second longitudinal end
[0060] **13** separator material
[0061] **14** anode
[0062] **14***a* anode material
[0063] **14***b* anode current collector foil
[0064] **16** cathode
[0065] **16***a* cathode current collector foil
[0066] **16***b* cathode material
[0067] **18** solid electrolyte
[0068] **20** hole
[0069] **22** container
[0070] **24** fluid inlet
[0071] **26** fluid outlet
[0072] **28** flexible housing
[0073] **28***a* seam
[0074] **29***a* internal shell
[0075] **29***b* external shell
[0076] **30** can
[0077] **30***a* open end
[0078] **32** lid
[0079] **32***a* recessed sections
[0080] **32***b* raised sections
[0081] **34** tubular section
[0082] **34***a* inner circumferential surface
[0083] **36** radially extending section
[0084] **40** tubular space
[0085] **42** internal retainer
[0086] **44** central hole
[0087] **46** axially extending centering part
[0088] **48** axially extending centering part
[0089] **49** axially extending centering part
[0090] **50** first lead
[0091] **52** fluid flow channels
[0092] **54** second lead
[0093] **56** electrically conductive ring
[0094] **58** flexible seal
[0095] **100** battery assembly
[0096] **160** parallel connection
[0097] **200** system
[0098] **202** gas inlet line
[0099] **204** gas outlet line
[0100] **206** controller
[0101] **208** control valve
[0102] **210** pressure relief valve
[0103] **212** check valve
[0104] **214** HVAC system
[0105] **216** evaporator
[0106] **218** compressor
[0107] **220** condenser
[0108] **222** dryer
[0109] **224** expansion valve

What is claimed is:

1. A battery comprising:

a tubular battery cell including an anode, a cathode and a solid electrolyte, the battery cell defining a central through hole; and

a container, the battery cell being inside of the container, the container including a fluid inlet configured to provide fluid to the central through hole and a fluid outlet configured to receive fluid from the central through hole.

2. The battery as recited in claim **1** wherein the battery cell has a jelly roll configuration.

3. The battery as recited in claim **1** further comprising a flexible housing inside the container surrounding the battery cell, the flexible housing configured for allowing the battery cell to radially expand and contract within the flexible housing.

4. The battery as recited in claim **3**, wherein the flexible housing is a bag including an internal polypropylene shell formed of a first polymer and an external shell formed of a second polymer.

5. The battery as recited in claim **3**, wherein the flexible housing is oversized with respect to the battery cell to allow for up to 10% expansion of the battery cell during charge and/or discharge of the battery cell.

6. The battery as recited in claim **1** wherein an outer circumferential surface of the battery cell and an inner circumferential surface of a tubular section of the container define a tubular space therebetween to allow gas to flow outside of the outer circumferential surface of the battery cell to generate a radially inward force on the outer circumferential surface of the battery cell.

7. The battery as recited in claim **6** further comprising an internal retainer inside the container and fixed to a first longitudinal end of the battery cell, the internal retainer including a central hole therein aligned with the central through hole of the battery cell to cause fluid flowing from the fluid inlet to the fluid outlet to pass through the central hole of the internal retainer.

8. The battery as recited in claim **7** wherein the container includes a can and a lid fixed to the can, the can including the tubular section, the internal retainer being physically connected to the lid and held in place within the can by the lid.

9. The battery as recited in claim **8** wherein a radially extending section of the can defines a first longitudinal end of the container and the lid defines a second longitudinal end of the container, the lid including the fluid outlet, the radially extending section of the can including the fluid inlet.

10. The battery as recited in claim **8** wherein the radially extending section of the can includes an axially extending centering part protruding into the central through hole of the battery cell at a first longitudinal end of the battery cell.

11. The battery as recited in claim **10** wherein the internal retainer includes an axially extending centering part protruding into the central through hole of the battery cell at a second longitudinal end of the battery cell.

12. The battery as recited in claim **8** wherein the internal retainer is electrically connected to the lid and electrically connects a first lead of the battery cell to the lid.

13. The battery as recited in claim **12** wherein the can is electrically connected to a second lead of the battery cell.

14. The battery as recited in claim **13** wherein the first lead is positive and the second lead is negative.

15. The battery as recited in claim **8** wherein the lid is configured to provide fluid flow channels between the lid and the internal retainer for fluid to pass from the central through hole of the battery cell to the tubular space.

16. A battery assembly comprising:

at least two of batteries as recited in claim **1**, at least one of the batteries being an upstream battery with respect to a fluid flow and at least one of the batteries being a downstream battery with respect to the fluid flow; and

the central through holes of the at least upstream battery and the at least one downstream battery being aligned with each other and the fluid inlet of one of the at least one downstream battery being connected to the fluid outlet of the at least upstream battery.

17. A battery system comprising:

the battery assembly as recited in claim **16**; and

a pressurization system for conveying compressed gas into the batteries to generate a force on the battery cells for maintaining uniform pressure on each battery cell.

18. A method of constructing a battery comprising:

inserting a tubular battery cell into an open end of a can such a tubular space is present between an outer circumferential surface of the battery cell and an inner circumferential surface of the can, a first longitudinal end of the battery cell facing a radially extending end section of the can, the radially extending end section of the can including a fluid inlet configured to provide fluid to a central through hole of the battery cell, the battery cell including an anode, a cathode and a solid electrolyte;

installing an internal retainer onto a second longitudinal end of the battery cell; and

connecting a lid to the open end of the can such that the lid is electrically connected to the internal retainer during operation of the battery, the lid including a fluid outlet configured to receive fluid from the central through hole.

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